SFU SIMON FRASER UNIVERSITY
ENGAGING THE WORLD

MSE220-Engineering Material Project Topic

*Submit your **team project report** together with your **EduPack project file** to "Canvas' Assignment" menu. One submission is required from one project team. The due date is **December 3rd, 2024 at 23:59 pm.***

Design of A Fan Blade for an Aircraft Turbine Engine*





General Electric GE90: the largest physical engine in aviation history!

**This project is a simplified version of a basic part of a large project by General Electric Aero Engines.*

BACKGROUND:

Most of the thrust for propulsion in a modern commercial jet engine comes from a large diameter fan at the front of the engine, which is driven by the low-pressure turbine at the rear of the engine. The fan, similar to a room fan, consists of multiple blades that rotate about the fan axis at high speed, and push the air backward past the engine[1]. The power to the fan is transmitted

[1] Check out this link: http://www.youtube.com/watch?v=TyMGLjo6Ojo

through a long shaft down the centerline of the engine and then into a large disk into which the fan blades are inserted.

To perform its aerodynamic function (pump air) the fan blade has to have a specified size and shape. Here, the size to be considered is 900 mm long x 300 mm wide by a maximum thickness of 20 mm. The shape is typically a complex airfoil with twist and taper along the radial axis, and may be hollow for weight reduction. For the sake of simplicity in this problem, assume all the blades to be rectangular in cross section with internal cavity as shown below:

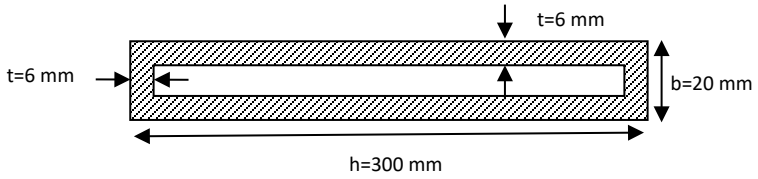


Figure 1. Cross section of the designed cavity inside fan blades (not scaled)

The mass of the blades rotating at high speed not only creates high stresses in the blades themselves, it also requires that the fan disk be strong enough to hold the blades, further increasing the weight of the system. Since increased engine weight increases the takeoff weight and reduces the payload, it is desired to minimize the weight of the fan blade, subject to bending stiffness of the blades.

Cost is always a constraint in jet engines, particularly commercial ones, and it is desired to keep blade cost as low as possible. Cost is not only a function of starting material, but also fabrication. To work out the fabrication costs, calculate the surface area, internal (inside cavity) and well as external, and multiply by $6000\ CAD/m^2$.

Blades are also experiencing a tensile stress (σ) caused by centrifugal loading. For a given angular velocity, assume that this stress scales just with the density ρ of the material: (here, the shape effect has already been considered in the coefficient below, good for you!)

$$\sigma = 0.05\,\rho\ \left(\sigma\ in\ MPa, \rho\ in \frac{kg}{m^3}\right)$$

A blade must withstand these loads for at least 100,000 engine flights (1 start-stop cycle per flight). You may simplify this fatigue criterion by assuming that a material will last **100,000 cycles** if the tensile stresses are held below one half of the ultimate tensile strength σ_{ts}.

Tolerance to damage (dents, cracks) from impact of foreign objects (rocks, birds) is also important. A 0.5 mm deep impact-induced crack should not propagate under the cyclic loads imposed by centrifugal force. Fast fracture will occur if the fracture toughness reaches the value

of $\sigma\sqrt{\pi a}$; use this criterion to set a lower limit for the allowable fracture toughness of your material ("a" is the value of impact-induced crack size in meter).

In addition to the centrifugal loads, blades are subjected to lateral vibrational loads due to air turbulence, system excitation, etc. The method used to reduce vibrational fatigue is to keep the natural frequency as high as possible. Natural frequencies of blades scale as $(E/\rho)^{1/2}$, so the answer is to make this as large as possible (or make ρ/E as low as possible)

The temperature of the fan blade will reach a maximum of 150°C during operation.

REQUIREMENTS:

Write a complete report containing necessary graphical illustrations, and fully explain your selection method step by step using Granta Edupack. Find the best three materials, according to your method, meet the objectives and constraints of the fan blades.

ECO-CRITERIA REQUIREMENTS:

Select three top-ranked candidates for the blades and compare them using the eco-criteria point of view. Assume that the blades are manufactured in a factory in Beijing, China and then shipped to Vancouver, Canada. The shipment of the blades can be through ground and/or sea, but not air. Other eco-audit properties required in Granta EduPack are listed in the following table.

Number of blades	1000
End of life	Landfill
Product life	3 years
Usage	250 days/year
Travelled distance	10000 km/day

Final Selection:

Compare top three materials and select the best candidate materials as your final selection based on 1- Eco-Criteria, 2- Total cost (material + manufacturing cost) and 3- Other factors (like keep mechanical properties in high temperature), and provide justification for this selection.

Tips:

- Set the units to SI and the price to Canadian Dollars.
- The second moment of area for the defined cross section is: $I = \frac{1}{6}h^3t(1 + 3\frac{b}{h})$ where t<<b, h (value of t is much less than b and h) if needed.
- Depending upon the chosen material, consider an appropriate primary process for the Eco Audit.